

Nabita Penmetsa

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RESEARCH FOCUS

I study the safety of agentic AI systems, focusing on multi-agent evaluation, organizational safety, and interaction-level alignment. My research combines empirical LLM evaluation (Inspect AI, RLHF/RLAIF, and constitutional AI) with economics, game theory and organization theory to understand how organizational structure, incentives, coordination, and information flow shape agent behavior and amplify risks such as cascading failures, hidden-objective propagation, and emergent coordination. My long-term research goal is to develop rigorous evaluation frameworks and alignment methods to improve the safety, reliability, and governance of increasingly autonomous AI systems.

RESEARCH INTERESTS

Multi-agent safety and organizational alignment · LLM evaluations and benchmarking · RLHF and preference optimization · Constitutional AI and role-based alignment · Governance of agentic systems · Incentive and mechanism design for AI

AI SAFETY RESEARCH

Secret Loyalties in Agentic Organizations

Multi-agent evaluation and organizational safety · Lead researcher · Inspect AI · Writeup available on request

Setup. A controlled evaluation of how a hidden objective, seeded in one compromised agent, propagates through multi-agent LLM organizations under varying delegation rules and information-sharing regimes.

Implementation. Built centralized and decentralized organizational scaffolds in Inspect AI, with structured transcript logging and custom safety metrics measuring blast radius, task-goal degradation, and hidden-goal completion.

Findings. Decentralization substantially reduces effective blast radius, while broad information-sharing amplifies hidden-objective spread and emergent coordination even when only one agent is compromised.

Contribution. Connects scalable-oversight and multi-agent safety research to a design-space perspective on how organizational structure and information flow shape cascading risk in agentic systems.

Interaction-Level Governance for Healthcare LLMs

Interaction-level alignment · Lead author · Under review at npj Digital Medicine (Nature)

Framework. A three-layer governance framework that stratifies each interaction by clinical risk, user state, and decision proximity rather than treating all queries symmetrically. Risk tiers map to response strategies (clarification, escalation, bounded non-directive guidance), addressing the “gray-area” failure mode where models oscillate between sycophancy and blanket refusal.

Uncertainty mechanism. Proposed inter-model agreement on clinical risk assessment across multiple LLMs as an uncertainty signal for conservative arbitration.

Ongoing. Investigating RLHF and preference-optimization approaches to operationalize interaction-level safety policies as post-training targets.

Constitutional Training for Role-Based Alignment

Interaction-level alignment · Ongoing research

Question. Can constitutional training mitigate harmful behavior arising from strong persona identification and persistent role context—role-play jailbreaks, sycophancy under persona pressure, goal drift within long conversations?

Approach. Designing an experimental framework based on the attached-vs-detached role-identification distinction, evaluating whether separating role-playing from goal pursuit improves alignment while preserving task performance. Comparing constitutional and preference-based training regimes across diverse personas.

Honesty–Sycophancy Tradeoff in LLMs with Evolving User Preferences

Incentives and trust in agentic systems · Working paper

A formal model of the strategic tension providers face between honesty (long-run trust) and sycophancy (short-run engagement) when users have limited ability to detect sycophancy and preferences drift with model output. Proposes

trust as a benchmark metric for alignment and governance evaluation, and identifies conditions under which honesty is the equilibrium strategy.

AI SAFETY TRAINING

BlueDot Impact — Technical AI Safety Course, 2026. RLHF, interpretability, evaluations, governance, frontier safety.

BlueDot Impact — AGI Strategy Course, 2026.

MLCommons — Member, AI Risk and AI Safety & Science Working Groups (2023–present).

TECHNICAL SKILLS

ML & Alignment. Inspect AI, multi-agent agentic evaluations, RLHF and RLAIIF, DPO, constitutional AI, prompt engineering, evaluation pipeline design.

Programming. Python (primary), scientific Python stack (NumPy, pandas), API integration with frontier LLMs.

Modeling. Game theory, mechanism design, incentive analysis, formal modeling of multi-agent systems.

PUBLICATIONS AND RESEARCH (ECONOMICS OF DIGITAL PLATFORMS)

Gal-Or, E., Gal-Or, R., & Penmetsa, N. (2018). “The Role of User Privacy Concerns in Shaping Competition between Platforms.” *Information Systems Research*.

Gal-Or, E., Gal-Or, R., & Penmetsa, N. (2018). “Can platform competition support market segmentation? Network externalities versus matching efficiency in equity crowdfunding markets.” *Journal of Economics and Management Strategy*.

Gal-Or, E., May, J. H., & Penmetsa, N. (2015). “Dynamic Pricing of New Services in Subscription Markets.” *Production and Operations Management*.

Working Papers Relevant to AI Safety

Penmetsa & Anand. *Organization of Platform Markets: Coordination, Decision Rights and Information*. Revising for resubmission, *Management Science*. Formal analysis of centralized vs. decentralized coordination under agent competition and information asymmetry — directly extended in “Secret Loyalties.”

Penmetsa, Anand, & Goyal. *The Ethical Flywheel: Responsible Platforms and Thriving Markets*. Dynamic model of ethical vs. profit-maximizing platforms under bounded rationality; relevant to reputation systems, incentive design, and alignment of platform-mediated agents.

EDUCATION

Ph.D., Management Science, University of Pittsburgh (Katz Graduate School of Business), 2014.

M.B.A., Indian Institute of Technology, Madras, 2007.

B.E., Mechanical Engineering, Osmania University, 2005.

ACADEMIC POSITION

Assistant Professor of Operations and Information Systems, David Eccles School of Business, University of Utah, 2014–present.

SELECTED RECENT TALKS (AI & DIGITAL PLATFORMS)

“Ethics and Platform Responsibility,” INFORMS Annual Meeting, Seattle, 2024.

“Ethics and Platforms,” POMS Annual Conference, Minneapolis, 2024.

“Organization of Platform Markets: Coordination, Decision Rights and Information,” Theory in Economics of Information Systems Workshop, San Francisco, 2023; INFORMS, Phoenix, 2023; CIST, Phoenix, 2023.

PROFESSIONAL SERVICE

Program Chair, Conference on Information Systems and Technology — *Amplified Intelligence: Economic, Work, and Societal Implications*, 2025. Reviewer for *Management Science*, *Information Systems Research*, *MSOM*, *JEMS*, *POM*, and *Decision Sciences*.

AWARDS

Rockwell Foundation Fellowship, University of Pittsburgh (2010–11). Doctoral Fellowship, University of Pittsburgh (2010–14). Multiple Excellence Awards, Citigroup (2008–09).